EVA Laptop Bag

(08.03.0670)

I. Fabric

1. Material: 0.5mm single-sided black dustproof silicone fiberglass cloth (Color: Black)

2. Material Features: Silicone coating + fiberglass cloth substrate, combining dustproof sealing (smooth surface for easy cleaning), fire retardancy (no open flame propagation), wear and scratch resistance (surface silicone layer resists friction), uniform thickness (tolerance ≤ ±0.05mm)

3. Core Function: As the outer fabric of the EVA battery box, it provides dustproof protection and a fire barrier, meeting the environmental tolerance requirements of outdoor/industrial battery boxes.

4. Compliant Standards/Certifications: Fire resistant compliant with UL94 V0, RoHS IEC 62321, CE, REACH, PFAS Test, Prop 65 Test, Glowing Wire Test, Explosion-proof Test

II. Middle Layer

1. Material: EVA foam material

2. Core Function: Cushioning and shock absorption protection, utilizing the elasticity and resilience of EVA to absorb the impact force generated by external collisions and drops (e.g., 1.2m). In drop tests, the battery's stress is reduced by over 70%, preventing damage to the battery components. The pores of the EVA foam structure block heat transfer, and combined with the flame-retardant lining and fiberglass fabric outer layer, it enhances the battery compartment's temperature insulation, reducing the impact of external temperature differences on the battery.

Zipper: Fire-retardant metal zipper

1. Material Features: Solves the problem of weak fire resistance at the opening and closing parts of the bag, ensuring overall fire resistance without gaps; meets the durability requirements for frequent daily opening and closing.

2. Compliant Standards: The flame-retardant fabric tape meets UL94 V0 rating and RoHS test.

Fireproof document bag (shoulder strap version)

(08.05.0223)

Outer Layer 1：Liquid Silicone Fiberglass Cloth (Silicone Coating + Fiberglass Cloth Base)

1. Material Characteristics: Fire-retardant, waterproof, moisture-proof, wear-resistant, scratch-resistant, flexible and conforming (avoids scratching internal equipment).
2. Function: Inner layer protection to prevent damage from friction between the computer and the inside of the bag.
3. Standards: Corresponding standards/certifications: UL94 5VA (flame retardant), RoHS IEC 62321 (environmental hazardous substance control), CE (EU safety compliance), REACH (EU chemical compliance), PFAS Test (perfluorinated compound compliance), Prop 65 Test (California chemical safety), Glowing wire Test (glow wire flame retardancy test).

Middle Spacer：PP Board

1. Material Characteristics: Lightweight and high-strength, good compression and impact resistance, excellent formability.
2. Function: Provides core structural support for the bag, dispersing external forces such as compression and impact to maintain the bag's shape stability.
3. Standard: Compression resistance meets GB/T 1041-2008 Plastic Compression Performance Standard.

Accessories Materials: Fire-retardant Zipper

1. Material Characteristics: Solves the problem of weak fire resistance at the opening and closing parts of the bag, ensuring overall fire resistance without gaps; meets the durability requirements of frequent daily opening and closing, while adapting to the dual characteristics of "waterproof + fireproof" for the bag body.
2. Corresponding Standards: Flame retardant fabric tape meets UL94 V0 rating and RoHS test.

Hardware Safe Case

(08.05.0224)

I. Outer Layer Fabric

1. Material: 0.5mm single-sided black liquid silicone fiberglass cloth

2. Material Characteristics: Excellent fire resistance and flame retardancy; tough, wear-resistant, scratch-resistant, and anti-aging surface; excellent waterproof and moisture-proof performance; the silicone coating is tightly bonded to the fiberglass cloth, ensuring stable shape and resistance to deformation.

3. Core Function: As the outermost protective barrier of the battery box, it resists wear, water splashes, and minor impacts during daily use, while enhancing the overall fire resistance of the box and sealing external fireproof gaps.

4. Applicable Standards/Certifications: UL94 V0, RoHS IEC 62321, CE, REACH, PFAS Test, Prop 65 Test, Glowing Wire Test, IPX5

II. Middle Layer ： 15mm Thermal Insulation Fiberglass Insulation

1. Material: Flame-retardant fiberglass insulation cotton

2. Material Characteristics: Well-developed pore structure, excellent thermal insulation performance, effectively blocking the transmission of high and low temperatures; lightweight, soft, and elastic, with good cushioning and shock absorption capabilities.

3. Core Functions: Insulates the battery from the influence of external temperature differences, preventing damage to battery performance from high/low temperature environments; simultaneously absorbs external impact forces, assisting the outer layer material in dispersing external forces.

4. Corresponding Standards: Thermal insulation performance meets GB/T 10294-2008 "Determination of Steady-State Thermal Resistance and Related Properties of Thermal Insulation Materials" standard; fire resistance meets UL94 V0 level and above flame retardant requirements.

III. Inner Layer: Aluminum Foil Fiberglass Cloth (Aluminum Foil + Fiberglass Cloth Base)

1. Material Characteristics: Fiberglass cloth base material + aluminum foil composite layer, using a hot-pressing composite process; good high-temperature stability, excellent long-term operating temperature range -40℃~250℃.

2. Functions:

① Thermal Insulation: Forms a highly efficient thermal insulation layer, preventing the battery's operating heat from being conducted outwards and maintaining a stable internal temperature.

② Electromagnetic Shielding: The aluminum foil layer effectively shields the battery's electromagnetic waves, preventing interference with surrounding electronic equipment and meeting the electromagnetic compatibility requirements of automotive/industrial equipment.

③ Moisture and Mildew Protection: The aluminum foil layer completely isolates moisture, preventing battery components from oxidizing due to moisture and extending battery life.

④ Structural Support: Combines flexibility and rigidity, protecting the battery module from compression and deformation.

3. Standards:

UL94 5VA, RoHS, IEC 62321, CE, REACH, PFAS Test, Prop 65 Test, Glowing Wire Test, Explosion-proof Test

Accessories Materials: Fire-retardant Zipper

1. Material Characteristics: Solves the problem of weak fire resistance at the opening and closing parts of the bag,

ensuring overall fire resistance without gaps; meets the durability requirements of frequent daily opening and closing, while adapting to the dual characteristics of "waterproof + fireproof" for the bag body.

2. Corresponding Standards: Flame retardant fabric tape meets UL94 V0 rating and RoHS test.

Fireproof vertical document bag

(08.03.0671)

Material 1: Liquid silicone fiberglass cloth (silicone coating + fiberglass cloth base)

1. Material characteristics: Smooth surface, good flexibility, adaptable to complex curved surfaces, resistant to bending; excellent heat resistance, stable performance within a long-term operating temperature range of -40℃ to 200℃, instantaneous temperature up to 800℃.
2. Functions: ① Prevents external fire sources from contacting the battery, delaying the spread of fire; ② Waterproof and dustproof, with an IPX5 protection rating, preventing rainwater and dust from entering the battery; ③ Abrasion-resistant and tear-resistant, resisting friction and mechanical damage during transportation and installation; ④ Resistant to acid, alkali, oil, and other chemical substances, adaptable to complex working environments.
3. Applicable Standards/Certifications: UL94 V-0 (Flame Retardancy), RoHS IEC 62321 (Environmental Hazardous Substances Control), CE (EU Safety Compliance), REACH (EU Chemical Compliance), PFAS Test (Perfluorinated Compound Compliance), Prop 65 Test (California Chemical Safety), Glowing Wire Test (Glow Wire Flame Retardancy Test)

Material 2: Aluminum Foil Fiberglass Cloth (Aluminum Foil + Fiberglass Cloth Base)

1. Material Characteristics: Fiberglass cloth base material + aluminum foil composite layer, using a hot-pressing composite process; good high-temperature stability, excellent long-term operating temperature range -40℃~250℃.
2. Functions:
① Thermal Insulation: Forms a highly efficient thermal insulation layer, preventing the battery's operating heat from being conducted outwards and maintaining a stable internal temperature.
② Electromagnetic Shielding: The aluminum foil layer effectively shields the battery's electromagnetic waves, preventing interference with surrounding electronic equipment and meeting the electromagnetic compatibility requirements of automotive/industrial equipment.
③ Moisture and Mildew Protection: The aluminum foil layer completely isolates moisture, preventing battery components from oxidizing due to moisture and extending battery life.
④ Structural Support: Combines flexibility and rigidity, protecting the battery module from compression and deformation.
3. Standards:
UL94 5VA, RoHS, IEC 62321, CE, REACH, PFAS Test, Prop 65 Test, Glowing Wire Test, Explosion-proof Test

Accessories Materials: Fire-retardant Zipper
1. Material Characteristics: Solves the problem of weak fire resistance at the opening and closing parts of the bag, ensuring overall fire resistance without gaps; meets the durability requirements of frequent daily opening and closing, while adapting to the dual characteristics of "waterproof + fireproof" for the bag body.
2. Corresponding Standards: Flame retardant fabric tape meets UL94 V0 rating and RoHS test.

Fireproof Document Box

(08.06.0169)

I. Outer Layer Fabric

1. Material: 0.5mm single-sided black liquid silicone fiberglass cloth

2. Material Characteristics: Excellent fire resistance and flame retardancy; tough, wear-resistant, scratch-resistant, and anti-aging surface; excellent waterproof and moisture-proof performance; the silicone coating and fiberglass cloth are tightly bonded, ensuring stable shape and resistance to deformation.

3. Core Function: As the outermost protective barrier of the battery box, it resists wear, water splashes, and minor impacts during daily use, while enhancing the overall fire resistance of the box and sealing external fireproof gaps.

4. Applicable Standards/Certifications: UL94 V0, RoHS IEC 62321, CE, REACH, PFAS Test, Prop 65 Test, Glowing Wire Test

II. Middle Layer 1. 4mm Honeycomb Board

1. Material: Lightweight honeycomb structure board (usually polypropylene honeycomb core + composite substrate)

2. Material Characteristics: Honeycomb hollow structure, combining "lightweight" and "high strength" Features: Stable pressure and impact resistance, good formability and cost-effectiveness.

3. Core Function: Provides core structural support for the battery box, maintains the stability of the box shape, disperses external forces generated by external compression/collision, and prevents battery components from being squeezed and deformed.

III. Middle Layer 5mm Thermal Insulation Cotton (added to the top and sides)

1. Material: Flame-retardant thermal insulation cotton

2. Material Characteristics: Well-developed pore structure, excellent thermal insulation performance, effectively blocks the transmission of high and low temperatures; lightweight and soft texture with good elasticity, possessing good cushioning and shock absorption capabilities.

3. Core Function: (Top + four-sided directional protection) Isolates the influence of external temperature differences on the battery, preventing high/low temperature environments from damaging battery performance; at the same time, it absorbs external impact forces and assists the outer layer material in dispersing external forces.

4. Corresponding Standards: Thermal insulation performance meets GB/T 10294-2008 "Determination of Steady-State Thermal Resistance and Related Properties of Thermal Insulation Materials" standard; fire resistance meets UL94 V0 level and above flame retardant requirements.

IV. Inner Layer

Material : Aluminum Foil Fiberglass Cloth (Aluminum Foil + Fiberglass Cloth Base)

1. Material Characteristics: Fiberglass cloth base material + aluminum foil composite layer, using a hot-pressing composite process; good high-temperature stability, excellent long-term operating temperature range -40℃~250℃.

2. Functions:

① Thermal Insulation: Forms a highly efficient thermal insulation layer, preventing the battery's operating heat from being conducted outwards, maintaining stable internal temperature.

② Electromagnetic Shielding: The aluminum foil layer effectively shields the battery's electromagnetic waves, preventing interference with surrounding electronic equipment and meeting the electromagnetic compatibility requirements of automotive/industrial equipment.

③ Moisture and Mildew Proof: The aluminum foil layer completely isolates moisture, preventing battery components from becoming damp and oxidizing, extending battery life.

④ Structural Support: Combines flexibility and rigidity, protecting the battery module from compression and deformation.

3. Standards: UL94 5VA, RoHS IEC 62321, CE, REACH, PFAS Test, Prop 65 Test, Glowing wire Test, Explosion-proof Test,

Accessories Materials: Fire-retardant Zipper

1. Material Characteristics: Solves the problem of weak fire resistance at the opening and closing parts of the bag, ensuring overall fire resistance without gaps; meets the durability requirements of frequent daily opening and closing, while adapting to the dual characteristics of "waterproof + fireproof" for the bag body.

2. Corresponding Standards: Flame retardant fabric tape meets UL94 V0 rating and RoHS test.